

Team Name : Logic Builders

Team Members Names	Engineering Specialisation	Graduating Year	College Names
Meghma Das	Information Technology	2027	MCKV Institute of Engineering
Argha Pal	Information Technology	2027	MCKV Institute of Engineering

Tell Us a Bit About Yourself

	Meghma Das	Argha Pal
Any projects you've worked on	Movie Recommendation System, AI tutoring system for rural India, Meeting Scheduler Agent	Movie Recommendation System, AI tutoring system for rural India, Health Coach Agent
Past Hackathon experiences	Smart India Hackathon 2024, Smart India Hackathon 2025, Infosys Global Hackathon 2025, Brain Dead 2k26	Smart India Hackathon 2025, Brain Dead 2k26
Accolades or awards received	NA	NA
Other details (If any)	NA	NA

Problem Statement 3: Driving Behavior Score

Solution Overview

RideIQ is a real-time two-wheeler driving behavior scoring system using low-cost sensor fusion (IMU + GPS) and a jerk-based scoring algorithm.

Sensors: MPU-6050 IMU + NEO-6M GPS + Barometer on ESP32

Algorithm: Weighted jerk-based scoring across 5 parameters

Real-time: Score updated every 5 seconds via BLE/Wi-Fi

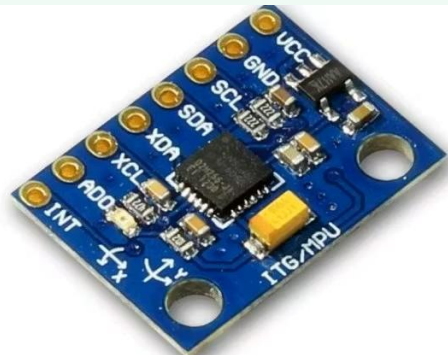
Cost: Total Bill Of Material under ₹1,000 per unit

Output: Dashboard, cloud log, insurance & fleet integration

Key Differentiators

- Context-aware scoring (speed zone + road type)
- Kalman filter IMU+GPS fusion for lean angle
- Jerk-based thresholds (mathematically grounded)
- Adaptive baseline learns rider over first 5 trips
- No internet required (edge inference on ESP32)

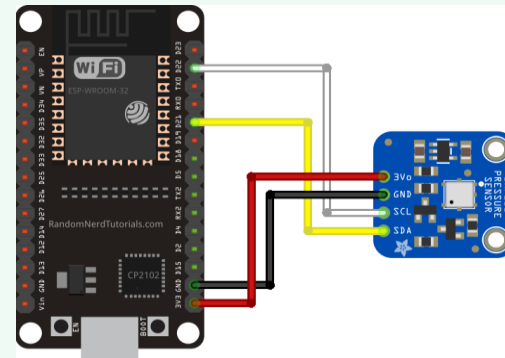
Sensors Details



A single chip combines three-axis accelerometer with three-axis gyroscope (6-axis) sensor. It uses I2C for communication via the address 0x68 (or 0x69 if the AD0 pin is set to true). It features an integrated Digital Motion Processor (DMP) that performs sensor fusion calculations to take the load off of the ESP32 and can provide quaternions directly as output. The IC runs on 3.3V to 5V power supply and can be connected to any of the 3.3V headers on the ESP32.



A u-blox GNSS receiver that provides output via NMEA 0183 sentences through the UART serial port with a default baud rate of 9600. It will provide approximately 2.5 m circular error probable (CEP) accuracy when there is an unobstructed view of the sky above. It requires a supply voltage of 3.3 V and a ceramic patch antenna (the antenna is typically soldered to it) oriented in a vertical position to receive signals from satellites to establish a position fix. Cold-start position fixes can take 30 to 60 seconds to establish.



When looking for a barometric altitude sensor, the BMP280 and MS5611 are two of the most common options. The BMP280 has a pressure sensing range between 300 and 1100 hPa and an altimetry resolution of approximately giving it a great general use sensor. The MS5611 has a wider range of pressure measurements and its altimetry resolution is significantly better than the BMP280 making it the preferable option for drones and higher precision applications. Both sensors can be connected to an MPU-6050 using I2C or SPI comms and also have a built-in temperature sensor in addition to measuring altitude through the use of the barometric formula. Because both sensors use unique I2C addresses, they can share I2C SDA and SCL lines with an MPU-6050 while all three sensors communicate on the same I2C network.

Design Details

Market Research & Literature Survey

- India has 300M+ registered two-wheelers; two-wheelers account for ~35% of all road fatalities (MoRTH 2023)
- Existing OBD-based systems (Mfine, Acko, Bajaj Allianz) work only for 4-wheelers; no affordable two-wheeler solution exists
- UBI (Usage-Based Insurance) market projected to reach \$28B globally by 2027 — two-wheelers are an untapped segment

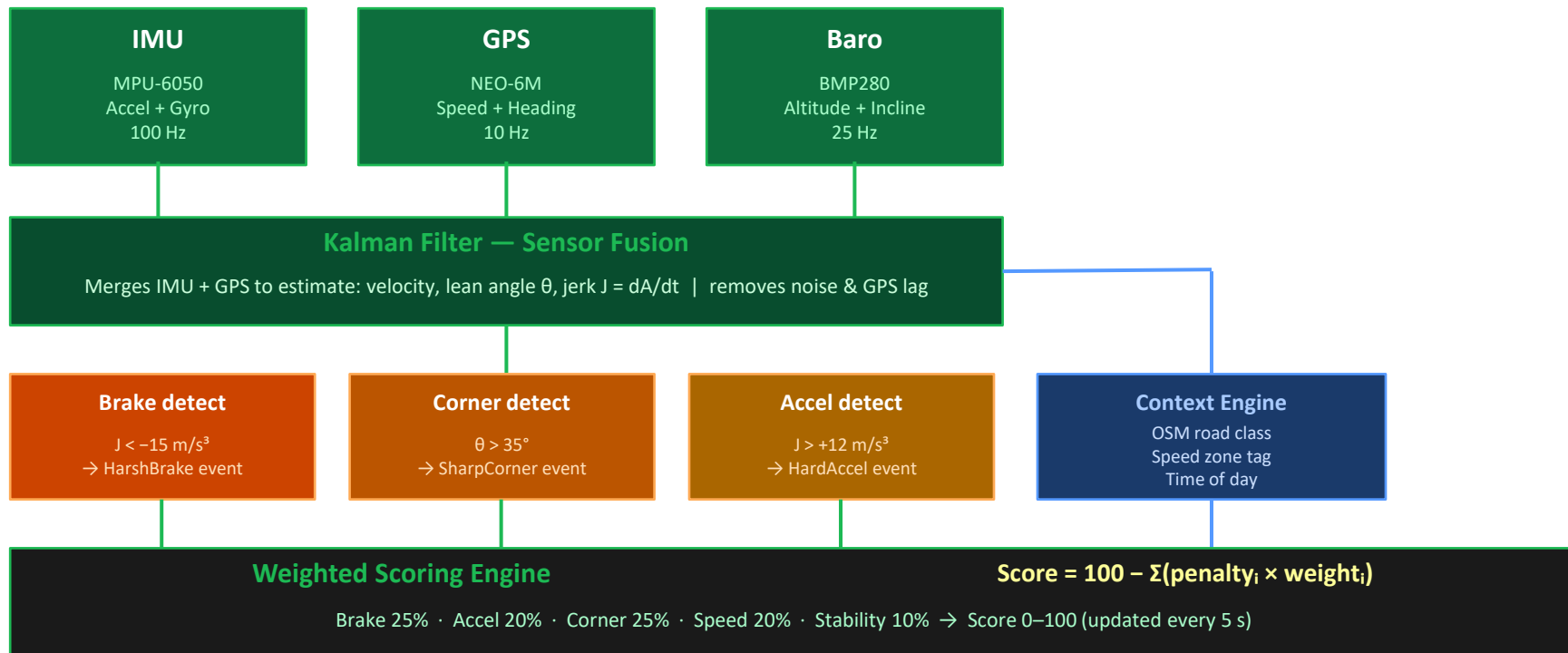
Working Principle & Scoring Formula

Score = 100 - $\sum (\text{penalty}_i \times \text{weight}_i)$ where jerk $J = dA/dt$ (m/s³) | lean angle $\theta = \int \text{gyro}_z dt - \text{GPS_heading_correction}$

Parameter	Trigger Threshold	Weight	Score Penalty
Harsh braking	$J < -15 \text{ m/s}^3$	25%	-10 per event
Hard acceleration	$J > +12 \text{ m/s}^3$	20%	-8 per event
Sharp cornering	$\theta > 35^\circ$	25%	-10 per event
Speed limit violation	>120% of limit	20%	-5 per km
Lane stability	Lateral jerk > 8	10%	-3 per event

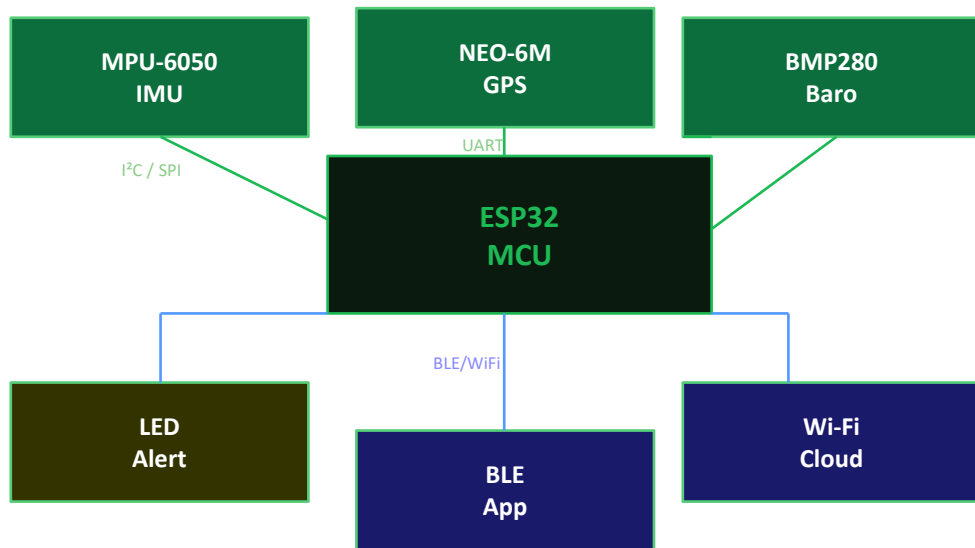
Design Details — Algorithm Logic Flow Diagram

End-to-end data flow: from raw sensor input to final driving behavior score



Design Details — System Architecture & Benchmarking

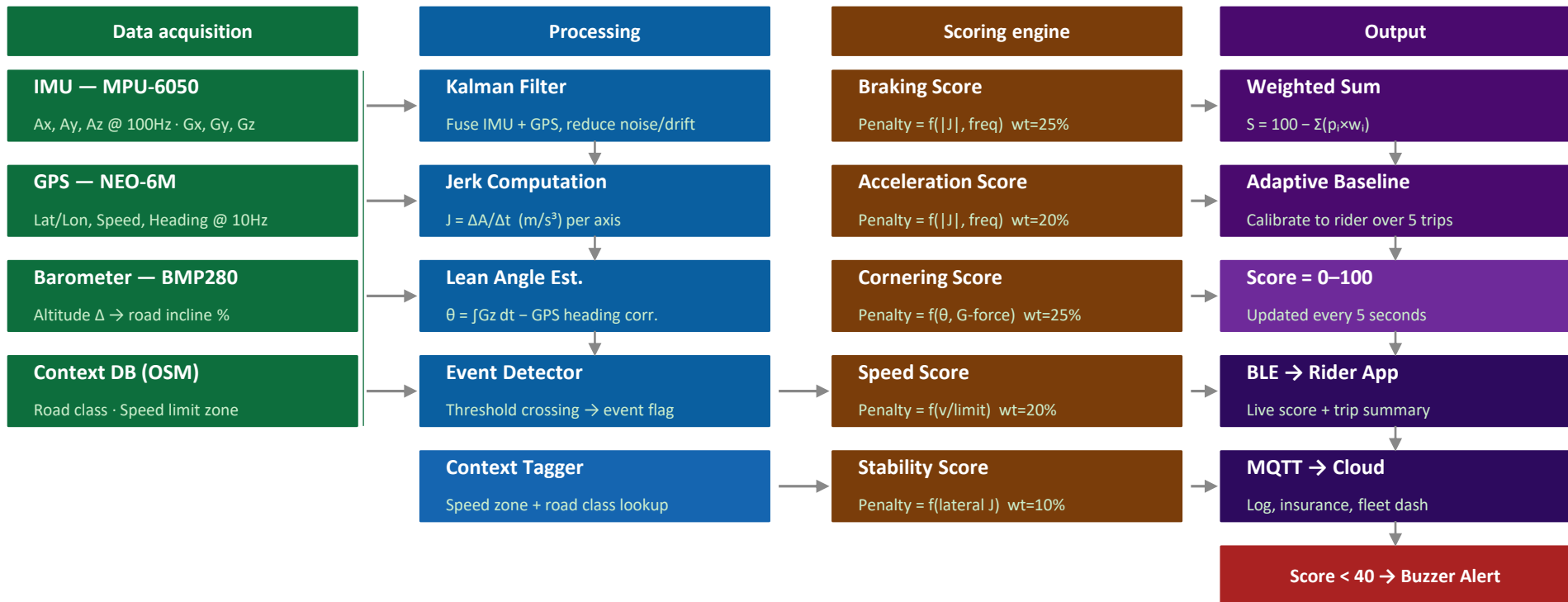
Hardware block diagram



Benchmarking vs existing solutions

Feature	RideIQ	Others
2-wheeler specific	✓	X
Works offline	✓	X
Lean angle tracking	✓	X
Cost < ₹1,000	✓	X
Real-time alert	✓	Partial
Context-aware scoring	✓	X
Jerk-based (not G-force)	✓	X

Logic Flow Diagram — Real-Time Scoring Pipeline



Major Risks & Mitigation

RISK: IMU drift over long rides

High

✓ Kalman filter with periodic GPS re-calibration every 60s

RISK: False positives on speed bumps / potholes

Medium

✓ Barometer + accelerometer Z-axis filter distinguishes road anomalies from braking

RISK: GPS signal loss in urban canyons

Medium

✓ Dead reckoning using IMU integration maintains scoring continuity for up to 30s

RISK: Battery drain on ESP32

Low

✓ Deep sleep mode between events; duty cycle reduces power to <80mA average

RISK: Rider behavior varies by region/road type

High

✓ Adaptive baseline + OSM-based road context engine normalizes scoring per road class

Business Potential

₹4,200 Cr+

India UBI market
(2026E)

300M+

Registered 2-wheelers
in India

35%

Road fatalities
from 2-wheelers

₹1,000

Hardware Bill Of Material
per unit

Revenue Model & Market Segments

B2B — Insurance Companies

UBI premium discounts drive rider adoption; insurers pay ₹50–100/month per device subscription

B2B — Fleet Operators (Zomato, Swiggy, Dunzo)

Reduce accident liability; driver score used for bonuses & training

B2C — Rider Safety App

Freemium app; premium tier at ₹99/month unlocks detailed trip analysis

B2G — Smart City / Traffic Dept

Aggregate anonymized data for hotspot analysis and road safety policy

Plan & Budget Needed

Prototype Development Milestones



Budget Required for Prototype

Item	Estimated Cost
Hardware (ESP32, IMU, GPS, barometer ×3 units)	₹3,500
PCB fabrication & enclosure (weatherproof)	₹4,000
Cloud hosting (AWS IoT / Firebase — 6 months)	₹2,500
App development (React Native)	₹0 (in-house)
Testing & field data collection	₹2,000
Contingency (10%)	₹1,200
TOTAL	₹13,200

VARROC *eureka* CHALLENGE

Thank You

Team Logic Builders | Problem Statement 3: Two-Wheeler Driving Behavior Score